

A woman with blonde hair tied back is climbing a dark, textured rock face. She is wearing a purple tank top, dark pants, and climbing shoes. She has a climbing harness with a helmet and other gear attached. A red rope is visible running down the side of the rock. The background shows some green foliage at the bottom.

CHAPTER 1

PHILOSOPHY OF TRAINING

PHOTO: JULIE ELLISON

PRINCIPLES OF TRAINING



AS TO METHODS THERE MAY BE A MILLION AND THEN SOME, BUT PRINCIPLES ARE FEW. THE MAN WHO GRASPS PRINCIPLES CAN SUCCESSFULLY SELECT HIS OWN METHODS. THE MAN WHO TRIES METHODS, IGNORING PRINCIPLES, IS SURE TO HAVE TROUBLE.”

RALPH WALDO EMERSON

Training is about teaching your body to perform better in an athletic environment. Humans are remarkably adaptable; you can get extremely flexible, extremely strong, or create extreme stamina, all based on what you ask your body to do in training. How do you ask? By following the rules.

Over the millenia, athletes and coaches have identified several principles of training which are the rules by which training occurs. Put simply, the principles are as follows:

OVERLOAD

You’ve got to ask your body to do more than it’s currently capable of in order to stimulate improvements. I frequently use the analogy of learning to read; unless you challenge yourself with harder and harder books, you’ll never get past *Green Eggs and Ham*. Overload

is the most basic of our training principles. This principle simply shows us that in order to adapt, we must subject ourselves to more. This can be in the form of overcoming very heavy resistance (strength), long duration under submaximal load (endurance), or any other facet of training.

One mistake many of us make is to look at overload only in terms of intensity: We lifted more, sent a higher grade, etc. A key factor in training, however, is one's capacity to do it at all. After years of hammering the lectern about how everyone should be doing supplemental strength training, I realize now that some people entering the sport lack the simple capacity to exercise at all for a normal week. Similarly, some climbers' training consists of a couple hours of bouldering 1-2 times per week. These climbers then get to the crag and start to fatigue after only a few pitches. A large part of my training philosophy now involves overloading not only an athlete's top-end values of strength and power, but also low-end capacity so that they can train harder, for longer, down the road.

SPECIFICITY

Our bodies adapt to exercise very specifically. If you want to get better at running, run. If you want to get better at jumping, jump. Although there is a great deal of crossover between certain sports and exercises, you're going to be most successful if you simulate your goal activity in your training. For our purposes, we will be training for climbing with an eye toward specific goals. These should be relative to your life, your sport, and your ambitions.

I tend to spend a lot of time discussing specificity, as it is easily misunderstood and poorly applied. Specificity can be movement-specific, energy system-specific, or both. There is a time to apply each of these in training. For example, indoor bouldering is very close to real rock climbing in its motor and metabolic demands,

so we'd consider it a highly specific training mode. Riding a dual-action bike could be very close to climbing's metabolic demand, but would not be considered motor-specific. Indoor bouldering is similar to the movement you'd do on real rock, but is different in metabolic demand to what you'd face at a sport crag.

As important as specific training is, let me underscore that not all training can be, nor should be, an exact mimic of the sport. There is a time and place for non-specific modes of training, especially in a skill-oriented endeavor like climbing. Due to the taxing nature of the sport (neurologically, logistically, and physically) optimum training requires that we include non-specific modes in order to optimize volume and intensity of effort.

“An athlete’s fitness should be improved without having to perform a high volume of exercises very similar to the competitive activity. Doing otherwise will leave no room for increasing sport specific training loads at the next stage and will close the possibility of further improvement.”

- Thomas Kurz, Science of Sports Training

PROGRESSION

At my home crag, the trail from the parking lot leads right to the steepest, tallest part of the cliff. Once you've reached the wall, you can choose to walk right to a shorter wall that features several 5.6 to 5.10 routes, walk left to a group of 5.11 climbs, or drop your pack right where you stand - at the bottom of several 5.12 and 5.13 routes. Over the years, it has become apparent to me that each climber's decision on where to climb on a given day determines where they can climb. Those who continually take a right get very good at climbing short 5.9s. Those who climb on the

steep wall above the trail tend to fall off a lot at first, but eventually get better because they are reaching outside their current abilities.

To continue to adapt, you must continue to increase the stress on your body. Related to the overload principle, a body will adapt to most training stresses after a short time, and thus must be regularly challenged by greater loads/durations/speeds. Load is not the only way to progress. If you can't add load, you can almost certainly add volume, increase speed, add more sessions, increase session density, etc.

INDIVIDUALITY

Group programming is where you make money in the fitness industry. Fill a room with warm bodies, have one instructor take the group through a hard session, and you have the roots of a profitable system. Profitable for the facility, but not optimal for the students. Even in a session that is designed to be “scaled” to several fitness levels, you'll see individuals that are under-stressed by the workout, and others that are so overworked that they can't possibly recover by the next session. Although motivation might be high, adaptation is sacrificed for most individuals.

We each adapt to stresses differently. When two friends start an exercise program at the same time, it's inevitable that one will progress more quickly. This doesn't make one athlete “right” and the other “wrong.” The only mistake is to not take your own adaptation into consideration when planning the next session. Failure to adjust your training to your own abilities will slow your progress...or give you a season-long injury.

Further, some athletes will adapt really well to specific loads. A naturally powerful athlete will gain power quickly while he might not gain capacity well at all. Athletes with great natural endurance will require longer training cycles to develop strength. Building training plans with this in mind will save you a lot of heartache.

REVERSIBILITY

Use it or lose it. The athlete's body is an expensive machine to maintain. The sorts of adaptations we make in training, such as building muscle mass or maintaining an artificially low level of body fat, are specialized adaptations that your body won't maintain unless it is continually reminded to do so. Like the tan on your back or the calluses on your hands, the changes your body makes will stick around as long as you keep up the constant exposure to overload that built them in the first place.

The body reverses the adaptation process for the sake of efficiency. This is reversibility. One of the main aims of *Logical Progression* is to mitigate this principle.

STRENGTH IS FUNDAMENTAL

The fitness quality known as strength is the basis for all of the facets of fitness that we rely on in climbing. The programs outlined in this book rely heavily on building a strong base. In 2014 I released the book *Strength: Foundational Training for Rock Climbing* wherein I lined out the reasons for building strength first-thing in a program. I refined this text in 2019 with Charlie Manganiello under the title *Unstoppable Force: Strength Training for Climbing*. Both books are aimed at addressing the need for general strength upon which to build athleticism. No matter what quality you are trying to develop as a rock climber, increasing maximum strength will increase those numbers, too.

There are many qualities one can develop, such as power, strength, or stamina. Each of these has a rate of decay that indicates how quickly it begins to detrain after you have "topped it off." The bright side of this decay is that the qualities that decay quickly can be gained back quickly. The ones that decay slowly tend to be hard

to develop. This is known as adaptation persistence. Strength and power are among the most persistent qualities you can develop. For this reason, strength and power training must be present year-round for your entire climbing career.



Adapted from "Practical Programming" (2007)

We train strength so that skills are easier to acquire.

We train strength so that you can recover more quickly from intense bouts of climbing.

We train strength so that every move you make on the rock and every step you take with a loaded pack are just that much easier.

Some athletes like strength training so much that it becomes a second passion for them. Although there is nothing wrong with this path, my recommendations will not waiver. The amount of supplemental training you do should always take a back seat to the practice of the sport. I highlighted this idea in Strength:

"We look to strength training in climbers not as a direct means to improvement, but as an adjunct to effective practice in the sport. This supplemental training can occasionally include weight training in a gym. However, one of the biggest misconceptions about weight training is the belief that it can directly improve your sport performance. Resistance training among top-level athletes in every sport is almost universal.

However, its use and applicability decline with highly skill-oriented sports. Climbing is just such a sport. The popularity of “elite fitness” programs, and even high-intensity resistance programs aimed directly at climbers only proves this point. If a general weight training program makes you better at climbing, you suck at climbing.”

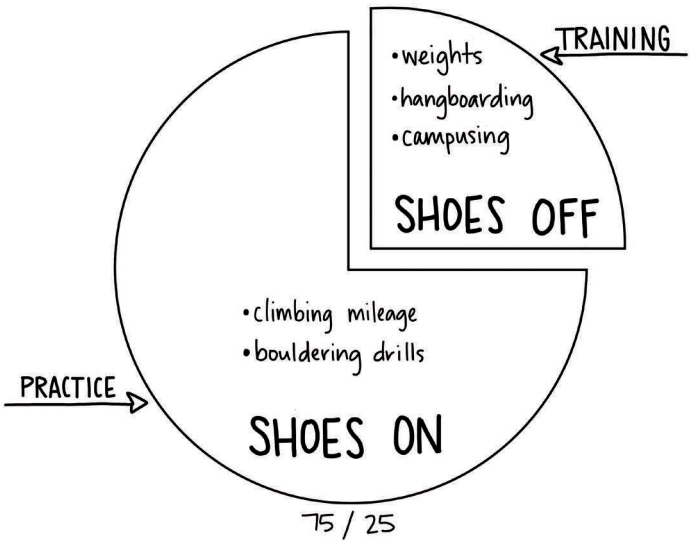
KEEPING TRAINING IN PERSPECTIVE

We are rock climbers, not runners, or weightlifters, or hangboarders. As I said before, it’s easy to get swept up in the training, and before long your rationalizations start to take over: “If I just get a bit stronger, I’ll be ready to start climbing again.” “My conditioning isn’t good enough for climbing right now...I’ll start again next month.”

I will remind you again that climbing is more an activity to be practiced than a sport to be trained for. In fact, the technical component is so high that one can attain advanced levels of the sport while being relatively weak and poorly conditioned. You cannot, however, attain high levels of the sport with massive strength and poor skills. With this in mind, I want to give you two reminders that will help keep training in perspective.

FIRST IS THE 75/25 RULE. Seventy-five percent of your “training” should actually be practicing the sport. Yes, I want you to hangboard and deadlift and campus, but you can never lose sight of the goal, which is to climb better. Much of your “training” will be bouldering in a gym, working specific drills, doing top rope laps or traverses, or redpointing mid-level routes, which are all to

be seen as practice. Then, about 25% of the time, all year, you'll train. This includes time on the training boards, in the weight room, and getting more mobile. A good rule of thumb: you'll spend 25% of the time training without your shoes, and 75% wearing them. The categorization is helpful; when you're practicing you are trying to get better, when you're training you're trying to build the fitness qualities that support better practice.



SECOND IS THIS: WHEN IN DOUBT, OPT FOR MORE CLIMBING. One of the great mistakes we can make is to drift away from climbing in favor of something where progress is more tangible. Moving toward a focus on campusing, to weight training, or even to such things as mountain running, are all paths I've seen good climbers take when getting better at climbing got hard.

There is no substitute for putting your head down and doing the work. There is no shortcut, no workaround. Understand that a plateau is a golden opportunity in your career to look a little deeper, to learn some better method for doing the sport. Relish it.

Training is what we do so that we can perform at a higher level in the sport of climbing. It must always be done with this question in mind: “How is this going to help my performance?”.

THE LONG VIEW OF STRENGTH

“There is no amount of work you could do today that will offset the progress you could have made in a properly structured week.”

- Christopher Sommer

A lot of getting better in climbing is simply making sure you’ve covered the bases physically: getting strong enough that you can have the opportunity to do high levels of practicing. When you’ve got a good physiological base, you can spend a lot of time trying hard moves. More than how far you can bury yourself in one session, it’s how long you can stay strong each season that determines progress.

TYPES OF STRENGTH

Depending on whose book you read, there are dozens of different “kinds” of strength. Perhaps the greatest treatment of the types of strength comes in Mel Siff’s seminal book *Supertraining*. Siff does a wonderful job of categorizing and explaining the myriad subtleties of strength. For climbing training purposes, we are interested in what he calls *relative strength*, *static (isometric) strength*, *strength-endurance*, and *rate of force production*.

Relative Strength is one's strength-to-weight ratio. Developing greater relative strength is critical to performance in sports such as gymnastics, wrestling, and climbing. Fighters also benefit greatly from increasing this facet of strength. This number can be manipulated by either gaining strength or by losing mass. For example, if you weigh 200 pounds and can do just a single pull-up, your relative pulling strength is 1:1. If you gained strength and lost weight, this would change. Let's say you dropped to 180 and could then do a pull-up with 40 pounds on your waist (180+40), your relative strength would take a significant leap. Naturally, relative strength applies to all of the derivatives of strength, too, such as power and endurance.

Isometric Strength is the ability to hold the muscles at a fixed length against force. This is especially useful in weightlifting, many strongman exercises, and is the Holy Grail of training for climbing. Almost all real-world applications of forearm strength heavily rely on isometric strength. In climbing we frequently combine high-load isometrics (holding on) with high load concentrics (moving) in adjacent muscle groups. Because of this, we need to pay special attention to integrating isometric and concentric strength in our programs.

Strength-Endurance is the ability to maintain a high level of force for extended periods. This is further subdivided into static strength endurance and dynamic strength endurance, as I mentioned above. If you simply held a large edge on a hangboard until you failed, this would be a measure of static strength endurance. Dynamic strength endurance would be displayed by repeating a movement pattern over and over again, as in high-rep pull-ups. Static strength-endurance (isometric endurance) is the far more common type in climbing, as we rarely fatigue the muscles responsible for dynamic movement more than we fatigue the forearms.



PHILOSOPHY OF TRAINING

PHOTO: MATT PINCUS

Rate of Force Production (RFP) In climbing training, this can be defined as training for maximum strength in minimum time. Climbers call this “contact strength,” or the ability to utilize a handhold as soon as the fingers touch it. Think of it as how fast your grip can go from “zero to sixty.” More and more climbers are leaving the terminology contact strength and calling this facet “finger power” (because of the dynamic nature of the movement) instead. Either term works, as long as you’re clear that they are synonymous.

MUSCULAR FACTORS RELATED TO STRENGTH

There are several factors related to an athlete’s ability to create a strong body. These factors are present independent of training type selected. I discuss these here simply as background to understand what’s happening as you get stronger.

HYPERTROPHY

Muscle fiber cross-sectional area is important in the ability to generate force. Athletes with highly developed muscles in the forearm will have an easier time increasing their grip strength. Increasing size in the forearms is difficult due to the high concentration of slow-twitch fibers typically found in these muscles. Lots of climbers get hung up on this - they get too concerned about their inability to add muscle mass to the forearms and aren’t concerned enough about developing strength. Hypertrophy comes at a cost anyway: the more you develop the muscle mass in the forearm, the less capillary density and mitochondrial density you’ll have, and thus you could potentially see a decrease in endurance.

I think it is important to accept that you may or may not gain mass in your forearms through training, and that there has never been a correlation shown between a bigger forearm and higher climbing performances.

FIBER TYPES

The human muscle is made up of several groups of muscle fibers. Although a detailed look at muscle anatomy is beyond the scope of this book, I'll highlight that there are three primary types of muscle fibers that we are concerned about.

- Slow Twitch (ST), which are very endurance-oriented, but relatively weak
- Fast Twitch Oxidative (FOG), which are powerful yet have some endurance potential
- Fast Twitch Glycolytic (FG), which are extremely powerful, but have very little endurance.

If you're one of the lucky few to have more than average fast-twitch fibers in your forearms, chances are you'll get strong fast. Otherwise, be ready for plenty of hard work. In most athletes, we'll see predominantly slow-twitch muscle in all muscles of the forearm. The big muscles of the body usually tell a different story. The lats, pecs, and quads are predominantly fast-twitch power muscles, and thus lack the endurance capacity of muscles such as the abdominals, calf muscles, and muscles of the forearm.

Of course your individual fiber profile is largely determined by your genetics, heritage, and early activity selection when you were young...none of which you can do anything about now. The one thing you can do from training is to improve the efficiency and economy of the fibers you have.

CENTRAL NERVOUS SYSTEM INNERVATION

The wiring of your muscle fibers is as important as fiber type. The better your innervation and the ability of the motor units to fire in the most efficient sequence, the stronger the muscle becomes. Most of your finger strength gains will come from improved

innervation and firing. Vladimir Zatsiorsky (Science and Practice of Strength Training, 1995) details three main modes of increasing force generation:

1. Recruitment, the gradation of total muscle force by the activation and deactivation of individual motor units.
2. Rate Coding, the change in firing rate of motor units
3. Synchronization, the activation of motor units in a more or less synchronized way.

All three are improved by increasing load or intensity. Endurance (long and moderate) sessions have a negligible effect on force generation.

ADAPTATION TO OVERLOAD

Believe it or not, there are people out there, called “non-responders,” who do not adapt to the stresses of training. The vast majority of us react to an overload by becoming ever-so-slightly stronger after a workout, and we can repeat the overload over and over until we reach ideal strength. The rate of adaptation varies from person to person, and is primarily determined by your genetics.

BIOMECHANICS

A huge part of how strong you can become is how you’re built. Short fingers, short tendon insertions, and thick bones all contribute to an elevated ability to develop force. Shorter climbers can generate force more quickly. Be sure to avoid using your body type as an excuse, but recognize that there might be certain facets of the sport that take longer to develop, based on your build. More than any other factor in climbing, I find that people use their “build” as an excuse. Too short, too tall, small hands, big feet, you name it. It is critical that we understand that there is no changing this stuff,

that for everything that you think your partner possesses as an advantage, there is a probable downside, and that your excuses themselves are holding you back way more than your negative ape index.

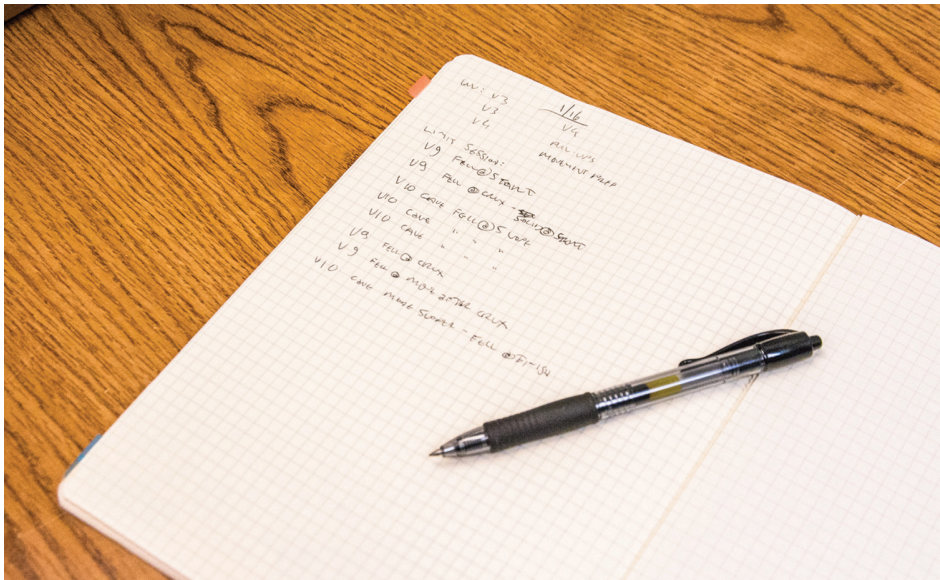
KEEPING TRACK OF TRAINING

When I first sit down with a new athlete, the first thing I ask the athlete is where they want to go. The question usually goes like this: “In six months, what do you want to have accomplished?” That part is easy for most of us - the dreaming is not hard to do. The next thing I want to know is how far the athlete has come in the last year. How many max-level redpoints? How many goal sends? How much has the athlete improved in the gym? How has his bodyweight been? It’s during this point in the interview that I get the blank stare. Most climbers can recall their hardest send in the last year, but they don’t really know why it happened.

A training log or journal is probably the most useful tool a climber can use in his training. By keeping a good log, you can precisely identify training patterns that worked and those that didn’t. It’s a little inconvenient to make notes during a workout, and it might make you feel a little uncool, but the value is worth it.

Training requires overload within the framework of both the session and the training cycle. If you don’t carefully plan overload session to session, you probably won’t see the progress you could. As I’ve stated before, just getting tired in the gym isn’t the answer. Pushing the loads the right amount higher and recovering correctly is how you really get better. The only way you can effectively do

this is to reference what happened last time, and the time before, and the time before. Most of us just can't remember that much.



What makes a good training log? Rule number one is to have one at all. Start with a calendar and write down the days you trained, what kind of training you did, and the total duration. That should take about five seconds a day. At the end of each month, you can compare total training time and intensity week-to-week and see if you're progressing things enough.

You can include more information each training session, and by doing so, gain more insight into what's really going on. The more information you gather, the better. In a weight training session, you'd keep track of exercises, sets, repetitions, load, and rest. Additionally, you could track tempo of movement, rest between exercises, and perceived exertion. With weights, it's easy to figure out how to progress - usually you just try to lift more.

In a climbing or bouldering session quantifying work is more difficult. Not all V3 problems are created equal! With this in mind, you'll want to pay attention to a few more details. In addition to writing the grade of each problem, I suggest tracking total

problems / routes climbed, angle of the route, type of moves / holds, and type of problem (i.e. techy, powerful, continuous, etc.). You'll also need to track performance. Did you flash it? Take three tries? Have you worked a problem for a few days?

More important than the numbers, perhaps, is tracking how often you had to "reach" in your session. As I wrote earlier, the ability to give high-level efforts consistently is a better predictor of high-level performance than strength or endurance or flexibility or shoe sponsor. I find it is useful for climbers to track and try to improve this factor over time. With each set in the weight room, with each boulder problem, with each burn at the crag, you should be able to go back to your log after the effort and make a note of how hard you tried. Did you go for it on a tough move, or reach down for the quickdraw? Did you really go hard to that crimp or did you give it a "meh" effort? Was your mind in doing great pull-ups or were you just counting reps?

In my own training log, these efforts are marked with a simple star. At the end of each week, I count up the stars - some weeks this takes depressingly little time - and put them in a chart in the back of the log. Every month, I'll add up the weekly totals and compare. The goal over time is to create more situations where I give good solid tries in the gym, with the ultimate goal of going hard during each effort at the crag, as one of my admitted limiters is not trying too hard unless I am pretty damn sure I'll succeed.

-

Without too much number crunching, you'll get a pretty clear picture of where you're strongest and how fit you currently are. Without good record keeping, this isn't possible.

Once you've created the habit of logging your daily sessions, it's time to add another layer to your record keeping. First, you'll want to track adherence to your plan. I talk a lot about this with climbers who are unhappy with their current plan. The line usually

goes “I have been doing x training plan for 2 months and it’s not working.” My first question is, “How much of the plan did you do?” Again, the blank stare...The crazy thing is that most of us don’t even consider that changing *this* session and shortening *that* exercise will eventually erode the plan.

A good way to track adherence is to detail all of your planned sessions on the calendar for 4 weeks, then match up the actual training with the calendar as you progress. If you’re continually coming up short, it’s clearly an issue of planning. You’re setting yourself up to fail each and every cycle, which plays havoc with your mind.

Another critical tool is to assess just how much training you’re getting each cycle. We’ll look at the logs every 4 weeks and write down training hours per week, the total time spent in any given intensity level, significant sends or personal records in the gym, average bodyweight, girth measurements, and any testing data collected that month. This is a good time to use a spreadsheet. For the athletes I coach and in my own training, we use a tabbed spreadsheet with columns for each month’s training, and a tab for each year. Over time, we get a useful and realistic picture of what that athlete can do, and what kind of training is best for them when approaching a time they want to perform well.

Sound like a lot? It is. It’s a lot of useful information that only takes minutes to record. If you’re not logging your training, you have everything to gain by writing a few things down. With a few weeks of dedicated practice, you’ll start to see patterns in your training. You’ll learn what works...and that is more useful than any template-driven training app will ever be.

As you move through the exercises recommended later in the book, you’re going to refer back to your log frequently. This is not an optional aside to your training, but rather a guide. A few seasons into your training career, and it will be the most valuable piece of gear you own. The log itself might not matter to you. There are

many versions of training logs available online, and a simple spiral notebook could do the job. I personally like the gridded notebooks from Moleskine or Leuchtturm1917. These high-quality books will easily put up with a season of being tossed into a pack, and make writing columns or charts simple.